

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

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Project Name

Medical Diagnosis Center by Service Portal (ServiceNow)

Document Type

Project Design Phase-II

Purpose

This document defines how diagnostic-service information moves through the ServiceNow solution and translates functional requirements into user stories with acceptance criteria, priority and release.

1. DATA FLOW DIAGRAMS

Diagnostic information enters through the Service Portal, variables are validated, a booking record is created, availability and approval rules are evaluated, notifications are sent, fulfilment is performed, and report/closure outcomes are made available for tracking.

1.1 DFD Level 0 – Context Diagram

The Medical Diagnosis Center system is treated as one process. Patients provide diagnostic request information and receive status/notification outputs. Approvers provide decisions where required, while laboratory/diagnostic teams receive fulfilment context and return service or report outcomes.

External Entity	Data Flow	System
Patient / Caregiver	Patient details, test selection, appointment details and supporting documents	Medical Diagnosis Center ServiceNow System
Medical / Lab Approver	Approval request and approval/rejection decision for applicable tests	Medical Diagnosis Center ServiceNow System
Laboratory / Diagnostic Team	Fulfilment task context, sample/service outcome and report result	Medical Diagnosis Center ServiceNow System
Administrator / Reception	Scheduling support, service administration and operational monitoring	Medical Diagnosis Center ServiceNow System

Context flow: Patient request → ServiceNow diagnostic service → validation / approval / fulfilment → appointment, status and report outcomes.

1.2 DFD Level 1 – Detailed Data Flow

No.	Process	Data Flow / Function
1	Capture / Configure Test Request	Collect patient and diagnostic-test information through the Service Portal and provide the submitted booking context to the workflow.
2	Get & Validate Test Variables	Retrieve selected test, date, time slot, urgency and other configured variables; validate mandatory information and booking rules.
3	Create Booking / Diagnostic Record	Create the structured appointment/booking record and map submitted values into configured fields; initialize status as Requested.
4	Validate Availability	Check slot capacity and prevent duplicate or overlapping bookings for the relevant resource, technician or equipment.
5	Request Approval	Route special or high-value diagnostic requests to the configured approver when required.

6	Evaluate Approval	Check the approval outcome and branch according to Approved, Rejected or applicable exception path.
7	Update Status & Notify	Update the diagnostic record and send the corresponding lifecycle notification to the patient and relevant staff.
8	Fulfilment / Sample Collection	Create or assign the diagnostic task to the appropriate team and record sample collection or service completion.
9	Processing & Report	Track processing, complete the diagnostic report and make the report available through the configured patient experience.
10	Closure / Tracking	Maintain appointment, report and audit history for tracking, cancellation/rescheduling and operational reporting.

1.3 DATA STORES

Data Store	Purpose
D1 - Patient / Profile	Patient identity, contact and profile information required for diagnostic services.
D2 - Diagnostic Test Catalog	Available diagnostic tests, categories, fees, preparation information and service rules.
D3 - Appointment / Booking	Patient, selected test, date/time, status, slot/resource and booking history.
D4 - Diagnostic Report	Report metadata/status and digital report reference or attachment.
D5 - Payment / Invoice (Phase-based)	Payment status, fee, invoice and transaction information when payment is enabled.
D6 - Audit / Notification	Lifecycle notifications, status history and audit evidence.

2. USER STORIES

The user stories translate the Medical Diagnosis Center requirements into user-centered tasks. Priority and release distinguish core booking/workflow capabilities from later enhancements.

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Patient	Test Booking	MDC-1	As a patient, I can open the diagnostic service catalog and select the required test.	The selected test details and booking form open successfully.	High	Phase 1
Patient	Patient Registration	MDC-2	As a patient, I can register and maintain my profile information through the portal.	Required patient details are captured and the profile is available for booking.	High	Phase 1
Patient	Appointment Details	MDC-3	As a patient, I can enter the required appointment date, time slot and test information.	Mandatory booking variables can be entered and validated successfully.	High	Phase 1
System	Availability Validation	MDC-4	As the system, I can validate diagnostic slot availability before creating the booking.	Unavailable or conflicting slots are rejected and valid slots can continue.	High	Phase 1
System	Duplicate Booking Prevention	MDC-5	As the system, I can prevent duplicate or overlapping appointments for the same patient/resource.	Conflicting booking attempts are blocked according to configured rules.	High	Phase 1
System	Booking Record Creation	MDC-6	As the system, I can create a structured diagnostic booking record from submitted portal data.	The booking record is created with correctly mapped fields and Requested status.	High	Phase 1
Approver	Special Test Approval	MDC-7	As an approver, I can review a special or high-value diagnostic request and provide a decision.	The configured approver receives the request and the decision is recorded.	High	Phase 2
System	Approval Routing	MDC-8	As the system, I can evaluate the approval result and follow the correct workflow branch.	Approved requests continue; rejected requests follow the configured rejection path.	High	Phase 2
User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Patient	Booking	MDC-9	As a patient, I receive	A confirmation notification	High	Phase 2

	Notification		a notification when my diagnostic booking is confirmed.	is generated with appointment information.		
Lab Technician	Fulfilment Task	MDC-10	As a lab technician, I can access the assigned diagnostic task with complete patient and test context.	The assigned task contains the information needed for fulfilment.	High	Phase 2
Lab Technician	Sample Collection	MDC-11	As a lab technician, I can update the request when the sample is collected or the diagnostic service begins.	The booking status changes to Sample Collected or the configured equivalent.	High	Phase 2
System	Processing Status	MDC-12	As the system, I can track diagnostic processing and completion status.	Status transitions are recorded and visible to authorized users.	High	Phase 3
Patient	Report Availability	MDC-13	As a patient, I can view or download my completed diagnostic report.	Only an authorized patient can access the available report.	High	Phase 3
Patient	Status Tracking	MDC-14	As a patient, I can understand the current status of my appointment and report.	The current lifecycle status is available through the Service Portal.	Medium	Phase 3
Patient	Cancellation / Rescheduling	MDC-15	As a patient, I can cancel or reschedule an appointment according to configured rules.	Only permitted cancellations/rescheduling are accepted and history is updated.	Medium	Future
Administrator	Access Control	MDC-16	As an administrator, I can configure roles and ACLs to protect patient and diagnostic data.	Unauthorized users cannot access protected records beyond their permitted role.	High	Phase 4
User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Workflow Monitoring	MDC-17	As an administrator, I can inspect Flow Designer execution details when a diagnostic workflow fails.	Execution details identify the failed action, condition or input/output involved.	High	Phase 4
Administrator	Operational Dashboard	MDC-18	As an administrator, I can view diagnostic service metrics such as bookings, demand and turnaround time.	Configured dashboards show operational metrics for authorized users.	Medium	Phase 4
Patient	Online Payment	MDC-19	As a patient, I can pay for a paid diagnostic test when the payment phase is enabled.	Payment status and receipt/invoice information are recorded.	Medium	Future
Administrator	Maintainability	MDC-20	As an administrator, I can maintain the core workflow through ServiceNow configuration and Flow Designer.	Core workflow logic can be understood and modified without unnecessary custom server-side code.	Medium	Phase 4

3. ACCEPTANCE FOCUS

Critical acceptance path: patient enters valid diagnostic information → selects a test → appointment variables are validated → slot availability is confirmed → booking record is created → approval is requested when required → approval is evaluated → status is updated → notification is sent → diagnostic fulfilment is completed → report becomes available → request remains available for tracking and closure.

Testing should focus on variable mapping, slot validation, duplicate booking prevention, record creation, approval routing, status transitions, notification triggers, role-based access and report availability.